PARTHIK PRAJAPATI

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SUMMARY

- Software Engineer with 4+ years of experience delivering reliable backend and full-stack applications using Python, FastAPI, React, REST APIs, and microservices. Experienced in designing enterprise applications, distributed services, data workflows, and scalable solutions across complex business environments.
- Strong hands-on experience with AWS, Kubernetes, Docker, Kafka, PostgreSQL, CI/CD, and distributed systems, along with AI/ML technologies including LLMs, RAG, LangChain, OpenAI APIs, NLP, vector embeddings, and semantic search. Proven ability to apply these technologies to intelligent applications, automation, and data-driven business solutions.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, C++
Backend & Architecture: FastAPI, Node.js, REST APIs, GraphQL, Microservices, Distributed Systems, Event-Driven Architecture, System Design
AI & Generative AI: LLMs, RAG, AI Agents, LangChain, LangGraph, OpenAI API, Hugging Face, Vector Embeddings, Semantic Search, Prompt Engineering
Cloud & DevOps: AWS, Azure, Docker, Kubernetes, Terraform, CI/CD, GitHub Actions, Jenkins
Databases & Data: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Apache Kafka, Apache Spark, ETL
Frontend: React.js, Next.js, React Native, HTML, CSS
Testing & Observability: PyTest, Integration Testing, Prometheus, Grafana, Datadog, Linux, Git
Engineering Practices: OOP, SDLC, Agile, Scrum, Problem Solving, Code Reviews, Version Control, Debugging, Technical Documentation, Responsive UI Development, Cloud-Native Development, Scalable Systems, JSON

EXPERIENCE

Datadog, USA | Software Engineer **Jan 2026 - Present**

- Architected and developed scalable Python/FastAPI microservices for high-volume telemetry and cloud observability workloads.
- Built RAG pipelines integrating traces, logs, service metadata, vector embeddings, and semantic search to accelerate incident investigation and troubleshooting.
- Integrated AI-agent workflows with backend services using OpenAI APIs and LangChain, with monitoring for model latency, token usage, errors, and execution traces.
- Optimized PostgreSQL, Redis, and DynamoDB access patterns to improve API performance and reliability across distributed services.
- Designed event-driven data pipelines using Kafka and Apache Spark for large-scale telemetry processing and downstream analytics.
- Containerized and deployed production services using Docker, Kubernetes, and AWS, supporting scalable and highly available cloud environments.
- Strengthened application reliability through PyTest, integration testing, structured logging, Prometheus, Grafana, and performance monitoring.
- Automated CI/CD and release workflows using GitHub Actions and Jenkins, improving deployment consistency and engineering productivity.
- Applied distributed-system design patterns, fault-tolerant architecture, and performance analysis to improve service resilience and operational stability.

Mphasis, India | Software Engineer **Jan 2021 - Jul 2024**

- Engineered Python, TypeScript, and Node.js services for enterprise analytics and data-driven business applications.
- Designed REST APIs and event-driven microservices supporting distributed workflows across multiple enterprise systems.
- Developed machine-learning workflows using Scikit-learn and XGBoost for classification, forecasting, customer segmentation, and predictive analytics.
- Built Kafka-based data pipelines for large-scale transaction processing and integration with downstream applications.
- Applied NLP techniques for document classification, information extraction, and automated processing of unstructured business documents.
- Prototyped RAG-based document search using vector embeddings and semantic retrieval to improve enterprise knowledge discovery.
- Optimized PostgreSQL, MySQL, and MongoDB data models and queries for scalable application and reporting workloads.
- Containerized applications using Docker and Kubernetes and supported deployments across AWS environments.
- Automated CI/CD workflows using Jenkins and Git while implementing PyTest, integration testing, code reviews, and Linux-based troubleshooting.
- Applied Generative AI and prompt-engineering techniques to internal knowledge-retrieval and document-processing workflows.

EDUCATION

Master of Science in Computer Science
Montclair State University, NJ, United States
Bachelor of Engineering in Computer Engineering
Sai Institute of Technology & Engineering Research, India

PROJECTS

AI PDF Study Assistant (RAG-Based Document Intelligence)

Tech Stack: *Python, LangChain, OpenAI API, ChromaDB, Streamlit, FAISS, Hugging Face*

- Developed a Retrieval-Augmented Generation (RAG) application that allows users to upload PDF documents and ask natural language questions.
- Implemented document parsing, text chunking, embeddings, and semantic search to retrieve relevant information before generating AI-powered responses.
- Built an interactive Streamlit chat interface for accurate, context-aware document analysis.
- Designed the solution for document-intensive workflows such as insurance claims processing, enabling faster document review and reducing manual effort.

Intelligent Compliance & Laboratory Decision Platform

Tech Stack: *Python, FastAPI, PostgreSQL, React, LLMs, RAG, AI Agents, OCR, Vector DB, Docker, AWS*

- Architected an AI-assisted compliance platform that converts purchase orders, specifications, job orders, and laboratory reports into structured requirements and automatically performs cross-document test verification and result validation.
- Designed a hybrid reasoning engine combining deterministic compliance rules, OCR, LLM-based extraction, RAG, confidence scoring, and human-in-the-loop review to handle ambiguous technical documents while maintaining traceable evidence.
- Engineered asynchronous document-processing pipelines and an AI compliance copilot for conversational investigation, requirement traceability, exception analysis, and scalable processing of multi-document laboratory jobs.

AI-Native Real Estate Operations & Brokerage Intelligence Platform

Tech Stack: *Python, Django, PostgreSQL, React Native, Redis, LLMs, RAG, AI Agents, REST APIs, Docker, AWS*

- Architected a multi-user brokerage operations platform centralizing agents, leads, clients, properties, tasks, documents, communications, and role-based workflows across real-estate teams.
- Built Python/Django services and event-driven workflows for intelligent lead assignment, agent workload management, follow-up automation, property matching, and operational activity tracking.
- Developed a context-aware AI operations copilot using RAG and tool-enabled agents to reason across brokerage data, answer business queries, automate repetitive workflows, and surface actionable recommendations to agents and managers.

Autonomous Application Intelligence & Digital Twin Engine

Tech Stack: *Python, FastAPI, Next.js, PostgreSQL, LLMs, Multi-Agent Systems, RAG, Vector DB, Redis, Docker, Kubernetes, AWS*

- Architecting an autonomous platform that explores live web applications and constructs a continuously queryable application model capturing pages, navigation paths, workflows, functionality, dependencies, and observed behavior.
- Designed an agentic orchestration layer with specialized discovery, workflow-analysis, UX, technical-analysis, and validation agents that collaborate through shared state and evidence-backed reasoning rather than isolated LLM prompts.
- Engineering distributed analysis pipelines with persistent context, semantic retrieval, task queues, caching, observability, and containerized cloud services to support concurrent application analysis, failure recovery, and AI-generated technical recommendations.